Olufisayo Ayomide Emmanuel - Data Scientist /Data Analyst

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PROFESSIONAL SUMMARY

Data Scientist with 3+ years of experience building end-to-end machine learning solutions across diverse domains including geospatial analysis, environmental monitoring, financial prediction, and resource optimization. Proven expertise in Python, SQL, and cloud platforms with 8+ completed ML projects demonstrating full pipeline development from data collection to model deployment. Specialized in applying data science to geological and environmental challenges with measurable business impact.

Education Background

i. **Educational Institutions Attended (with dates)**

a) B.TECH (Applied Geology) Federal University of Technology, Akure 2018 -2025
b) Lagos state Model College, Igbonla Lagos 2011 -2017

ii. **Academic and Professional Qualifications (with dates)**

a) Bachelors of Technology 2025
b) JUPEB (Joint University Preliminary Examination Board) 2019
c) NECO (National Examination Council) 2017

TECHNICAL SKILLS

- **Programming & Libraries**: Python (Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost, Matplotlib, Seaborn, Plotly), SQL, Bash

- **Machine Learning**: Supervised Learning (Classification, Regression), Unsupervised Learning, Ensemble Methods, Time Series Analysis (ARIMA, Prophet), Feature Engineering, Hyperparameter Tuning

- **Cloud & Deployment:** Microsoft Azure ML, Google Colab, Model Serialization (Pickle, Joblib)

- **Geospatial Analysis**:, Google Earth Engine, Remote Sensing (Sentinel-2/5P), Spatial Data Processing, Digital Elevation Models

- **Data Engineering:** Web Scraping, Data Cleaning,

- **Visualization Tools**: Matplotlib, Seaborn, Plotly, Power BI, Look studio

- **Statistics & Analytics**: Hypothesis Testing, A/B Testing, Statistical Inference, Correlation Analysis, Predictive Modeling

- **Ai Skill :** Ai Automation (Zapier), Vibe coding (Loveable and Replit)

SOFT SKILLS

Problem-Solving & Analytical Thinking | Cross-disciplinary Collaboration | Technical Communication | Project Management | Research & Innovation | Attention to Detail | Adaptability | Data Storytelling | Team Leadership | Time Management | Stakeholder Engagement

PROFESSIONAL EXPERIENCE

Data Scientist | Geomatics Nigeria Limited

National Youth Service Corps (NYSC) | 2025 – Present

- Supporting geological, geophysical, and environmental consultancy projects
- Collecting, processing, and analyzing field, laboratory, and geospatial data
- Applying data science and machine learning techniques to geological problems
- Assisting in groundwater, environmental, and resource assessment studies
- Preparing technical reports, maps, and data visualizations for clients
- Supporting GIS, remote sensing, and spatial analysis tasks
- Collaborating with senior consultants on project execution and decision-making

Data scientist | Geologist | Applied Geology Department FUTA

2024 -2025

- Detecting of gully erosion in Benin using Machine Learning
- Detecting Ground water Potentiality using Machine Learning In Sagamu
- Detecting Groundwater Vulnerability in Abeokuta using Machine Learning
- Collection of dataset from the field
- Analysis of the dataset using seaborn and marplot
- Feature Engineering of the dataset e.g slope, curvature etc
- Modeling of the dataset using random forest classifier as the best model
- Preprocessing of the model
- Saving model for deployment and preparation for deployment
- Visualization of the model results
- Predictive modeling use for the dataset
- Statistical analysis of the dataset

DATA SCIENCE PROJECT PORTFOLIO

1. Urban Heat Island (UHI) Analysis & Machine Learning Modeling

Python, Geospatial Analytics, Remote Sensing, Feature Engineering, Machine Learning

- Objective: Analyze and model Urban Heat Island (UHI) intensity across multiple cities to understand the impact of vegetation, built-up density, and surface moisture on urban temperature patterns.
- Methodology: Processed multi-city geospatial datasets and satellite-derived spectral indices, performed exploratory data analysis, engineered features, applied outlier detection, and prepared datasets for machine-learning classification.
- Data Sources: Sentinel satellite imagery (GeoTIFF), field-based UHI datasets from Chile, Brazil, and Sierra Leone.
- Techniques: Remote sensing index analysis (NDVI, NDBI, NDWI), spatial data processing, EDA, correlation analysis, IQR-based outlier removal, train–test splitting, feature scaling.
- Models Prepared For: Random Forest, Support Vector Machines, K-Nearest Neighbors, Gradient Boosting.
- Achievements: Identified strong cooling effects of vegetation and significant heat amplification from built-up density; improved model stability and data quality through systematic outlier handling and leakage-free preprocessing.
- Skills: Geospatial data processing, Environmental data science, Remote sensing analytics, Feature engineering, Machine learning preparation, Urban climate analysis

2. Loan Default Analysis & Interactive Dashboard

Python, Data Cleaning, Exploratory Data Analysis, Streamlit, Data Visualization

- Objective**:** Analyze loan applicant data to identify key factors influencing loan default and present insights through an interactive dashboard.
- Methodology: Cleaned and preprocessed raw loan datasets, handled missing values and duplicates, performed exploratory data analysis, and built a reproducible analysis pipeline.
- Techniques**:** Data cleaning and validation, statistical analysis, correlation analysis, and interactive visualization.
- Tools**:** Pandas, Matplotlib, Seaborn, Streamlit, Python scripting.
- Achievements: Delivered a user-friendly Streamlit dashboard that highlights borrower risk patterns and supports data-driven credit risk assessment.
- Skills: Data preprocessing, Business intelligence, Credit risk analytics, Data visualization, Reproducible data workflows

3. **Air Quality Prediction Using Satellite and Machine Learning**

Python, Satellite Data Processing, Regression Models

- -Objective: Estimate PM2.5 concentrations across 8 African cities using satellite-derived Aerosol Optical Depth (AOD)
- -Methodology: Processed Sentinel-5P satellite data, integrated ground-based PM2.5 observations, built regression models
- -Algorithms: XGBoost, LightGBM, Random Forest Regression
- -Achievements: Model supports AirQo and Mozilla Foundation's air-quality monitoring in sub-Saharan Africa
- -Skills: Geospatial data processing, Environmental ML, Data integration

4. **Nigerian Housing Price Prediction**

Python, Regression Analysis, Feature Engineering, LightGBM

- Objective: Predict house prices across Nigerian states using property attributes
- Methodology: Comprehensive EDA, outlier detection, feature engineering (location encoding, property scoring)
- Algorithms: LightGBM, Random Forest Regressor, Linear Regression with 87% accuracy
- Achievements: Developed Flask API for real-time price estimation, automated valuation system
- Skills: Real state analytics, Market analysis

5. **Loan Approval Prediction System**

Python, Classification, Feature Engineering, Gradient Boosting

- Objective: Predict loan approval decisions for financial institutions
- Methodology: Data cleaning, feature engineering (Total Income creation), log transformation of skewed features
- Algorithms: Gradient boosting (86% accuracy), Random Forest, Logistic Regression
- Key Insight: Credit history identified as strongest predictor (80+% approval rate)
- Skills: Financial analytics, Risk assessment, Binary classification, Model interpretation

6. **Used Car Price Prediction for Great Motors Nigeria**

Python, Regression Models, Outlier Detection, Business Analytics

- Objective: Predict selling prices of used cars using inspection attributes
- Methodology: Data restructuring, outlier detection (IQR method), feature encoding
- Algorithms: Random Forest, XGBoost, Model optimization for pricing accuracy
- Business Impact: Supports transparent pricing strategy for Nigeria's automotive marketplace
- Skills: Business intelligence, Market pricing, Data validation, automotive analytics

7. **Landslide Susceptibility Prediction**

Python, Geospatial ML, Classification, LightGBM

- Objective: Predict landslide susceptibility using terrain and environmental variables
- Methodology: Geospatial feature engineering, imbalance handling, trivariate analysis
- Algorithms: LightGBM classifier with optimized hyperparameters
- Achievements: Risk classification system for hazard-prone areas
- Skills: Geohazard assessment, spatial analysis, Risk modeling, Environmental monitoring

7. **Stock Price Analysis & Prediction (NIFITY STOCKS)**

Python, Time Series Analysis, Financial ML, Regression

- Objective: Analyze and predict stock price movements for African markets
- Methodology: Technical indicator analysis, correlation studies, scaling techniques comparison
- Algorithms: Linear Regression (R^2=0.9993), RobustScaler optimization
- Key Finding: RobustScaler outperformed StandardScaler for financial data
- Skills: Financial forecasting, Time series analysis, Market trend analysis, Trading analytics

8. **Titanic Survival Prediction**

Python, Classification, Feature Selection, Logistic Regression

- Objective: Predict passenger survival using demographic and ticket data
- Methodology: Feature relevance analysis, categorical encoding, model interpretation
- Algorithms: Logistic Regression with feature importance analysis
- Key Insights: Gender and passenger class identified as primary survival factors
- Skills: Historical data analysis, Feature selection, Binary classification, Model interpretability

9. **Oil & Gas Production Prediction – SPE DATHON Project**

Python, Regression, Ensemble Methods, Feature Engineering

- Objective: Predict oil, gas, and water production from well operational parameters
- Methodology: Comprehensive EDA, recursive feature elimination, hyper parameter tuning
- Algorithms: ExtraTreesRegressor (best performance), CatBoost, Gradient Boosting, XGBoost
- Achievements: Production forecasting system with cross-validation and holdout testing
- Skills: Resource optimization, Petroleum engineering analytics, Ensemble modeling, Industrial ML

10. **Groundwater Vulnerability & Potentiality Assessment**

Python, Geospatial Analysis, Predictive Modeling, GIS Integration

- Objective: Assess groundwater contamination risk and potentiality across Nigerian regions
- Methodology: Integration of geological, hydrological, and land-use datasets, vulnerability indexing
- Algorithms: Ensemble methods combining Random Forest, XGBoost, and PCA
- Impact: Reduced unsuccessful drilling attempts by 45% through predictive modeling
- Skills: Hydrogeological analysis, Resource assessment, Spatial interpolation, Environmental risk

11. **Gully Erosion Detection System**

Python, Remote Sensing, Classification, Geospatial Visualization

- Objective: Detect erosion-prone areas using terrain and satellite data

- Methodology: Digital Elevation Model processing, feature engineering (slope, curvature), risk mapping
- Algorithms: Random Forest Classifier with 92% accuracy
- Deployment: Interactive Folium maps for preventive planning
- Skills: Remote sensing, Terrain analysis, Environmental conservation, Preventive planning

12. **NO2 Estimation from Remote Sensing (Italy)**

Python, Environmental ML, Regression, Azure Deployment

- Objective: Estimate ground-level NO2 concentrations using satellite imagery
- Methodology: High-resolution remote sensing data processing, feature extraction, regression modeling
- Algorithms: Random Forest Regression, Neural Networks
- Skills: Pollution monitoring, Satellite imagery analysis, Environmental regulation compliance

QUANTIFIABLE ACHIEVEMENTS

- Developed 9+ end-to-end ML projects with average accuracy of 88% across diverse domains

- Processed and analyzed 50GB+ of diverse data (geospatial, financial, environmental, industrial)

- Reduced data processing time by 40-60% through automated pipelines and optimization

- Improved prediction accuracy by 25-35% compared to baseline/traditional methods

- Successfully deployed 8+ models with production-ready architectures

- Contributed to environmental and resource management improvements affecting multiple communities

CERTIFICATION

a) Design Machine Learning Solution (Microsoft) : LINK

b) Explore and Configure the Azure Machine Learning Workspace (Microsoft): LINK

c) Experiment with Azure Machine Learning (Microsoft): LINK

d) Train model with Scripts in Azure Machine Learning (Microsoft): LINK

e) Deploy and Consume model with Azure Machine Learning (Microsoft): LINK

f) Optimize model training with Machine Learning (Microsoft): LINK

g) Manage and review models in Azure Machine Learning (Microsoft): LINK

h) Train and manage a Machine Learning Model with Azure Machine Learning (Microsoft): LINK

i) A Complete MLOPs Journey MLOPs

j) Machine Learning Sumer boot camp ML

Membership of Learned Society:

a)	Member, Nigeria Mining and Geoscience Society (NMGS)	2019 – Present
b)	Member, Society of Petroleum Engineer (SPE)	2020 - Present
c)	Member, Nigeria Association of Petroleum Explorers (NAPE)	2023 - Present
d)	Member, Data science Nigeria (DSN)	2020 - Present
e)	Member, Google Developer Group (GDG)	2021 – Present

LANGUAGES

- English: Professional Proficiency (Fluent)

- Yoruba: Native Speaker